

NCIE-012 Skeletal v0.1 → Full v1.0

Expansion Verification Report

UX/UI & Design System Specification — IN DEVELOPMENT / FOR HUMAN REVIEW OF FULL EXPANSION

1. Scope Verification

NCIE-012 v1.0 is a controlled full expansion of the approved NCIE-012 Skeletal v0.1: every approved chapter purpose, Salient Things to Note, principal component, upstream dependency/authority and Human Review focus is elaborated into UX/UI, workspace/panel, ARGUS presentation, interaction-state, accessibility and Design System architecture, without inventing a logo, brand palette, typography family, icon brand, imagery style, breakpoints, viewport minima, performance targets, accessibility conformance level, session timings, notification severities, usability targets, analytics thresholds, or mandatory design-system/analytics/prototyping/mapping/charting/accessibility-tool products.

2. 37-Chapter Integrity Verification

A programmatic scan of the final document confirms exactly 37 sequential 'NCIE-012 — Chapter N Expansion' labels, N = 1..37, each present exactly once, in the approved order, with no chapter added, removed, merged, split or renamed relative to the approved Skeletal v0.1 structure.

3. Salient Things to Note Hard-Gate Verification

Per directive item 79 — continuing the discipline established after the NCIE-009 Chapter-27 omission — every chapter must retain an explicit '<N>.0 Salient Things to Note' subsection. This build enforced the requirement structurally: the chapter-rendering helper (lib.py's add_chapter function) raises a build-time assertion error if a chapter's salient content is omitted, so the document cannot be assembled with a missing subsection. A programmatic scan of the final assembled document, rerun independently rather than inferred from the build script succeeding, confirms an explicit Salient Things to Note heading present for all 37 chapters (N = 1 through 37), with zero missing.

4. Skeleton Fidelity Verification

Each of the 37 chapters was authored directly from its approved skeletal Chapter Purpose, Salient Things to Note, Principal Components, Upstream Dependencies/Authority and Human Review Focus (see the accompanying Change Register for the per-chapter mapping). No approved chapter concept was omitted, and none was conceptually inverted.

5. No Brand, Threshold or Product Invention

A full-text scan for a named logo, brand palette, typography family, icon brand or imagery style found none invented (directive item 8) — Chapter 7 explicitly defers brand/visual identity inputs to Human Review (HR12-7-1). A separate scan for breakpoints, viewport minima, performance targets, accessibility conformance level, session timings, notification severities, usability targets or analytics thresholds found none invented (directive item 9) — every such matter is instead recorded as a named Human Review item (e.g., HR12-6-1 for form factors, HR12-29-1 for accessibility level, HR12-31-1 for UX performance targets). A third scan for mandatory design-system, analytics, prototyping, mapping, charting or accessibility-tool products found none (directive item 10).

6. Governing UX/UI Invariants Verification

The 19 invariants proposed in the approved Skeletal v0.1 ('Proposed UX/UI Invariants') are restated unchanged in front matter §0.3, and are additionally cross-referenced — not restated — against the 12 NCIE-008 governance invariants, the 16 NCIE-009 security invariants, the 19 NCIE-010 Evidence/provenance/audit invariants, the 20 NCIE-011 integration invariants, and other upstream invariants already binding from NCIE-005/006/007. A full-text scan confirms every invariant is both stated once in §0.3 and applied by name in at least one operative chapter (e.g., VISIBLE ≠ AUTHORIZED in Ch.4; HIDDEN ≠ SECURE in Ch.19; VPF PASS ≠ HUMAN APPROVAL in Ch.15; COLOUR ≠ SOLE STATE INDICATOR in Ch.29; DESIGN TOKEN ≠ BUSINESS RULE in Ch.7, 33).

7. UX/Upstream Boundary Verification

Chapter 1 Table 1 fixes what NCIE-012 does not own (canonical data, ARGUS behavior, Memory/Context semantics, Agent/VPF semantics, institutional governance, security enforcement, Evidence/provenance meaning, API contracts, domain business logic — all NCIE-001–011's). Every consequential representation in Chapters 2–35 is expressed as a rendering of a named upstream state with the accountable holder marked 'to be assigned,' never as NCIE-012 inventing new institutional authority, security mechanisms, Evidence semantics or API contracts of its own (directive item 7).

8. Misleading-Success Audit

A full-text scan (directive item 68) for UI wording that would turn accepted/submitted/timeout/pending/outcome-unknown states into confirmed success found none — Chapter 23 Table 29 (Async/Outcome Matrix) fixes a distinct, non-collapsible label for each of the seven async states, and Chapter 24 confirms errors preserve their semantic distinctions rather than a generic success/failure banner.

9. Machine-Authority Audit

A full-text scan (directive item 69) for presentation that would make ARGUS, an Agent, a Gate/VPF disposition, AI confidence or multi-Agent consensus appear to be Human Approval/Decision authority found none — Chapter 2 Table 3 (UX State Matrix), Chapter 11 (ARGUS presentation), Chapter 14 (Agent presentation) and Chapter 15 Table 18 (ARGUS/Agent/VPF Presentation Matrix) each explicitly prohibit an approval-style visual for a machine state.

10. Security-Leakage Audit

Verified (directive item 70) that hidden/disabled/redacted/restricted states do not leak protected data/existence and do not substitute for server-side authorization: Chapter 17 §17.1 confirms DISABLED CONTROL ≠ AUTHORIZATION ENFORCEMENT, Chapter 17 Table 21 (Security-State Matrix) fixes a policy-leakage constraint per security state, and Chapter 19 requires no existence leakage through counts, labels or layout artifacts.

11. Evidence-Semantics Audit

Verified (directive item 71) that search results, Memory, AI/Agent output, VPF PASS, annotations and charts are not presented as Evidence/Findings/Decisions absent governed state: Chapter 20 §20.1 and Table 25 (Evidence-State Matrix) prohibit collapsing a search result into an Evidence card, Chapter 12 distinguishes Evidence/Memory/AI-analysis/Agent-output content classes, and Chapter 10 requires charts to preserve provenance rather than imply Evidence status.

12. Responsive-Semantic Audit

Verified (directive item 72) that reflow/compact/mobile presentation never removes or changes critical validation, security, provenance, warning or approval semantics: Chapter 6 §6.1 states RESPONSIVE LAYOUT ≠ SEMANTIC CHANGE and requires critical governance/security states to survive reflow unchanged at every supported form factor.

13. Accessibility Audit

Verified (directive item 73) that colour is never the sole state cue and all critical interaction patterns have non-pointer/non-visual alternatives as applicable: Chapter 29 Table 36 (Accessibility Matrix) fixes a non-colour cue and a keyboard/screen-reader requirement for every major component/pattern class, consistent with COLOUR ≠ SOLE STATE INDICATOR and ACCESSIBILITY ≠ OPTIONAL ENHANCEMENT.

14. UX State, Role/Workspace and ARGUS/Agent/VPF Matrix Verification

Chapter 2 Table 3 (UX State Matrix, directive item 58) maps 8 upstream state classes to permitted representation and prohibited misleading representation, each citing its owning chapter. Chapter 3 Table 5 (Role/Workspace Matrix, directive item 59) maps 7 persona classes to default UX needs and panels with a repeated authorization note. Chapter 15 Table 18 (ARGUS/Agent/VPF Presentation Matrix, directive item 60) maps 5 state classes to distinct visual treatment and a Human-Primary caution.

15. Security-State, Evidence-State, Async/Outcome and Accessibility Matrix Verification

Chapter 17 Table 21 (Security-State Matrix, directive item 61) maps 7 security states to UX behavior and a policy-leakage constraint. Chapter 20 Table 25 (Evidence-State Matrix, directive item 62) maps 8 Evidence/provenance states to UX treatment and a prohibited-collapse rule. Chapter 23 Table 29 (Async/Outcome Matrix, directive item 63) maps 7 async states to label, affordance and retry/cancel rule. Chapter 29 Table 36 (Accessibility Matrix, directive item 64) maps 6 component/pattern classes to keyboard, focus, semantic, non-colour, screen-reader and caption/reduced-motion requirements.

16. Test-Status Integrity Verification

Chapter 35 Table 43 specifies 8 test classes (usability, accessibility, keyboard/screen-reader, responsive, security-state, async/error, voice, regression), each traced to the chapter(s) it verifies. Every test is marked TEST SPECIFIED only; a full-text scan confirms no test anywhere in this document is marked TEST EXECUTED, TEST PASSED, REPRESENTATIVE-USER VALIDATION complete, or claims IMPLEMENTATION EVIDENCE AVAILABLE (directive item 67) — detailed test-case authoring, execution and representative-user validation are explicitly deferred to NCIE-016.

17. No Implementation Claims

A full-text scan confirms no UX pattern anywhere in this document is claimed implemented, accessibility-conformant, usability-validated or production-accepted merely because this v1.0 specifies it (directive item 75) — every RACI/control-ownership table marks its holder 'to be assigned,' Chapter 34 distinguishes prototype approval from production acceptance, and Chapter 35 explicitly marks every test SPECIFIED, not executed or user-validated.

18. Human Review Register Mechanical Reconciliation (Correct From First Build)

Chapter 36's register was recomputed programmatically by scanning the Human Review Focus bullet text already rendered in Chapters 1-35, plus HR12-36-1 (Chapter 36's own item) and HR12-37-1 (Chapter 37's own item) — both defined once

and included in the register-construction list from the start of the build, a design choice made specifically to avoid repeating the register-reconciliation defect found and corrected in the companion NCIE-008 v1.0 document's initial build. Result: 37 total items, 28 Blocking / 9 Non-Blocking, 0 duplicate Review IDs, 0 orphan Human Review items. Classification breakdown: 25 Institutional Confirmation Required, 6 Security/Sovereignty Confirmation Required (Chapters 13, 17, 18, 19, 24, 32), 1 Legal/Policy Confirmation Required (Chapter 29 accessibility standard), 3 Engineering/Infrastructure Discovery Required (Chapters 6, 9, 31), 1 Proposed Design Default (Chapter 7 brand inputs), 1 Resolved by Upstream Baseline (Chapter 37).

19. Upstream Human Review Deduplication (Directive Item 66)

HR12-1-1 (confirm NCIE-012 acceptance/sign-off authority) restates the identical institutional question NCIE-008 already opened as HR8-10-1 (acceptance authorities for each NCIE-series document and chapter-level sign-off); Chapter 36 Table 46 cross-references HR8-10-1 rather than treating HR12-1-1 as an independent open question. Every other NCIE-012 Human Review item was individually checked against the NCIE-008/009/011 Human Review Registers and found to concern a UX/design-specific decision that those documents do not themselves resolve — no other duplicate was found.

20. Blocking Scope Discipline Verification

Each Blocking item in the Chapter 36 register states precisely which brand choice, threshold, ownership assignment or acceptance authority it blocks, rather than implying the whole document or the whole approved upstream chain is blocked. Non-Blocking items (Chapters 2, 5, 8, 11, 12, 14, 26, 28, and the resolved Chapter 37 item) are explicitly marked as not gating the corresponding chapter's UX operation in the interim.

21. Traceability and Downstream Non-Preemption Verification

Chapter 37 Table 47 traces all 37 chapters to their primary upstream authority (approved NCIE-001-011 and/or the approved NCIE-012 Skeletal v0.1); no chapter states a UX requirement without an identifiable source. Table 48 confirms this document hands over domain-workflow UX patterns to NCIE-013, operational responsibilities to NCIE-015, and the 8 UX/accessibility test classes to NCIE-016 — without specifying their technical implementation, consistent with directive item 20 (downstream non-preemption).

22. Version, Document-Control and Running-Header Verification

Cover and Document Governance table confirmed reading: Edition 'In development — Version 1.0 (Full Expansion of Approved Skeletal v0.1)'; all nine upstream authority rows (NCIE-001 through NCIE-011) read their approved current versions; a 'Prior baseline' row records the approved Skeletal v0.1 as historically preserved; Status reads IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION. The running header — rendered from a copied NCIE-011 v1.0 template to inherit this series' visual styling, including the logo, centered cover text, page break before Document Governance, and non-splitting governance-table rows already corrected in that series — was corrected to 'NCIE-012 | UX/UI & DESIGN SYSTEM SPECIFICATION | VERSION 1.0' and the footer corrected from 'CONTROLLED INTEGRATION DOCUMENT' to 'CONTROLLED UX DOCUMENT' and from 'NCIE-011' to 'NCIE-012' before this document was rendered to PDF.

23. Full-Document Cross-Reference Scan

Every occurrence of 'NCIE-011' in the final document was individually inspected (24 occurrences outside Upstream/Trace blocks). Each was confirmed to be a legitimate API/integration citation (e.g., 'NCIE-011 Ch.4-5', 'NCIE-011 async-state owner', 'mirroring NCIE-011 Ch.25's error taxonomy') consistent with directive item 7's requirement that NCIE-012 consume, not redefine, NCIE-011 API contracts — none is a leftover donor-template branding error. The cover, header, footer and Document Governance 'Document' row were separately confirmed to read NCIE-012, not NCIE-011.

24. Character-Encoding Integrity Check

Following the same build pipeline used for the whole NCIE series (UTF-8 source fragments executed via a Python exec() call with an explicit encoding='utf-8' argument, a fix first required during the NCIE-008 build), a full-document scan of the final NCIE-012 document confirmed zero mojibake occurrences across all 736 paragraphs and 49 tables.

25. Visual and Structural Verification

The complete document was opened in Microsoft Word, its Table of Contents field (a live Word field, not static text) was updated and confirmed to populate with correct chapter/section titles and real page numbers, repaginated to 63 pages, saved, and rendered to PDF. Independent text extraction from the rendered PDF confirms the cover occupies page 1 alone and Document Governance begins on page 2 (the manual page break), with the running header and footer correctly reading NCIE-012 branding from page 2 onward. No table clipping, overlapping text, or broken table was observed across the seven consolidated matrices, the Chapter 36 Human Review Register tables, or the Chapter 37 traceability/handover tables.

26. Final Human Review Extraction (Authoritative)

Performed directly against the final specification file by independently scanning every paragraph and table cell for the pattern HR12-<chapter>-<n>: TOTAL IDS FOUND = 37. UNIQUE IDS = 37. DUPLICATES = 0. OPEN = 36. RESOLVED = 1 (HR12-37-1, Resolved by Upstream Baseline). BLOCKING = 28. NON-BLOCKING = 9. CLASSIFICATION TOTALS: Institutional Confirmation Required = 25; Security/Sovereignty Confirmation Required = 6; Legal/Policy Confirmation Required = 1; Engineering/Infrastructure Discovery Required = 3; Proposed Design Default = 1; Resolved by Upstream Baseline = 1. ORPHAN IDS = 0 — every one of the 37 unique IDs found anywhere in the document also appears as a row in the Chapter 36 Complete Register table, which itself contains exactly 37 rows with no duplicate Review ID.

27. Three-Artifact Consistency Check

The NCIE-012 v1.0 main document, this Verification Report and the accompanying Full-Expansion Change Register agree exactly on: Source Version v0.1 (Skeletal) → Target Version v1.0 (Full); Authority chain approved NCIE-001 v1.3 through approved NCIE-011 v1.0, plus the approved NCIE-012 Skeletal v0.1; Status IN DEVELOPMENT — FOR HUMAN REVIEW OF FULL EXPANSION (not Approved); 37 chapters preserved in order (0 missing, 0 duplicate, 0 added, 0 unauthorized renaming); Salient Things to Note hard gate = 37/37 explicit subsections present; Human Review Register TOTAL = 37, BLOCKING = 28, NON-BLOCKING = 9, classification totals as in §26 above; orphan Human Review IDs = 0; duplicate Human Review IDs = 0; UX/accessibility test-class status = 8 tests specified, none executed, passed or user-validated; UX/UI invariants = 19 NCIE-012-specific plus cross-referenced upstream invariants; downstream handover = NCIE-013/015/016 each receive a defined interface without technical preemption. No mismatch was found between the three artifacts.

28. Final Readiness Disposition

NCIE-012 — UX/UI & Design System Specification v1.0 — Full Expansion of Approved Skeletal v0.1 — is READY FOR HUMAN REVIEW. The approved v0.1 skeleton remains historically preserved as the expansion architecture; every proposed brand choice, threshold, product choice and ownership assignment deferred in Chapters 1-35 remains a proposal pending resolution of the Chapter 36 Human Review Register; every §35.1 test remains SPECIFIED, not executed or user-validated, pending NCIE-016. This document is NOT marked Approved.